

The Multinomial Coefficient as a Product of Binomials

The multinomial coefficient represents the number of ways to partition a set of n elements into N groups of sizes $C_1, C_2, \dots, C_N$, where $\sum_{i=1}^N C_i = n$. It is defined as:

$$\frac{n!}{C_1!C_2!\dots C_N!} = \binom{n}{C_1, C_2, \dots, C_N}$$

Successive Selection Derivation

This can be derived by viewing the partition as a sequence of N successive selections. According to the **Product Rule**, the total number of ways is the product of the ways to choose each group from the remaining elements:

$$\binom{n}{C_1} \binom{n-C_1}{C_2} \binom{n-C_1-C_2}{C_3} \dots \binom{C_N}{C_N} \quad (1)$$

Algebraic Proof (Telescoping Product)

Expanding each binomial coefficient into its factorial form, we get:

$$\begin{aligned} \text{Product} &= \frac{n!}{C_1!(n-C_1)!} \cdot \frac{(n-C_1)!}{C_2!(n-C_1-C_2)!} \cdot \frac{(n-C_1-C_2)!}{C_3!(n-C_1-C_2-C_3)!} \dots \frac{C_N!}{C_N!0!} \\ &= \frac{n!}{C_1!(n-C_1)!} \cdot \frac{(n-C_1)!}{C_2!(n-C_1-C_2)!} \cdot \frac{(n-C_1-C_2)!}{C_3!\dots} \dots \frac{C_N!}{C_N!1} \\ &= \frac{n!}{C_1!C_2!C_3!\dots C_N!} \end{aligned}$$

Note: To use the `\cancel` command shown above, you will need to add `\usepackage{cancel}` to your LaTeX preamble.